Silicon Photonics Course Final Report

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1. Introduction

This report summarizes the layout and analysis work completed as part of the “Silicon Photonics Design, Fabrication and Data Analysis” course offered on edX by the University of British Columbia (UBC). It presents the photonic layout designed as the course project, implemented in KLayout using the EBeam PDK, and the subsequent simulation and measurement analysis performed on the fabricated devices.

The design includes two device sets: a set of Mach–Zehnder Interferometers (MZIs) for free-spectral-range (FSR) and group-index characterization, and a set of ring resonators (MRRs) spanning a radius/gap parameter sweep, used to study coupling regime and resonator performance across a fabrication-relevant design space. This report presents the complete layout, simulation, and measurement analysis for the MZI set; the ring resonator (MRR) analysis follows the same methodology and is presented separately.

2. Layout Description

The layout was implemented using KLayout with the EBeam PDK and submitted as two separate GDS files: EBeam_khaledmohammed_MZI.gds and EBeam_khaledmohammed_MRR.gds. All devices operate in the fundamental TE mode at 1550 nm.

2.1 Mach–Zehnder Interferometer Set

The MZI layout (Figure 1) contains six unbalanced Mach–Zehnder Interferometers (MZI1–MZI6) and two straight reference waveguides (StraightWG1, StraightWG2), arranged in two identical rows of four devices.

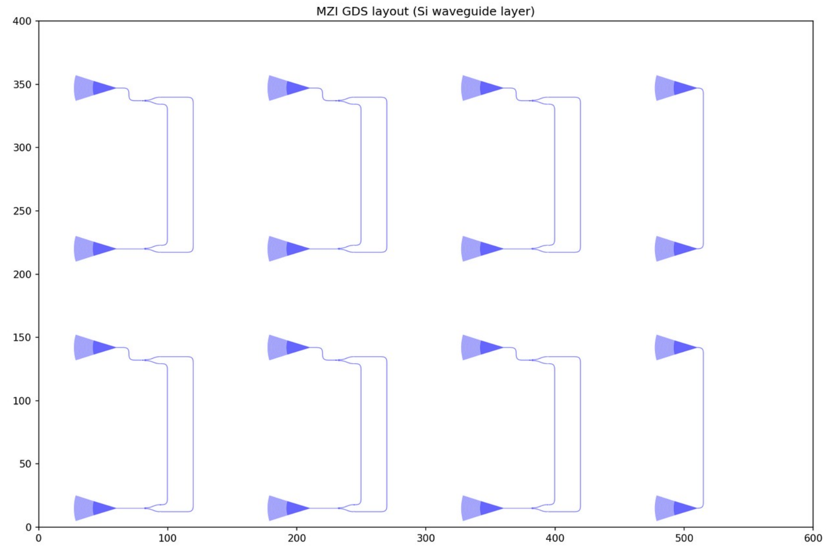

Figure 1: KLayout MZI design layout (Si waveguide layer), showing MZI1–MZI6 and StraightWG1/2.

Each MZI arm length was confirmed directly from the Lumerical/SPICE netlist generated by the KLayout-INTERCONNECT integration (Root_Element.spi), which lists the exact wg_length parameter for every waveguide component in the circuit. The two arms of each MZI are the waveguides connecting Y-branch1's two outputs directly to Y-branch2's two inputs; the shorter waveguides connecting the grating couplers to the Y-branches are routing only and are not part of the interferometer path. All six MZIs share the same nominal path-length difference, functioning as replicate devices for measurement redundancy rather than a design sweep:

Device ID	**Arm 1 (μm)**	**Arm 2 (μm)**	**Path difference ΔL (μm)**
MZI1	118.397	169.397	51.000
MZI2	118.397	169.397	51.000
MZI3	118.397	169.397	51.000
MZI4	118.397	169.397	51.000
MZI5	118.397	169.397	51.000
MZI6	118.397	169.397	51.000

Table 1: MZI arm lengths and path-length differences, confirmed from the SPICE netlist.

The two straight reference waveguides (StraightWG1, StraightWG2) each measure 133.897 μm and are used as the loss/de-embedding reference for the MZI spectra, following the cutback and de-embedding approach described in Section 5.

2.2 Ring Resonator (MRR) Set (to be presented separately)

The MRR layout (Figure 2) contains six ring resonators, spanning two ring radii and three coupling gaps, forming a 2×3 corner-style sweep of the coupling regime rather than a single fixed design. Full simulation and measurement analysis of this set follows the same methodology as Sections 4-6 below and is presented in a follow-up submission.

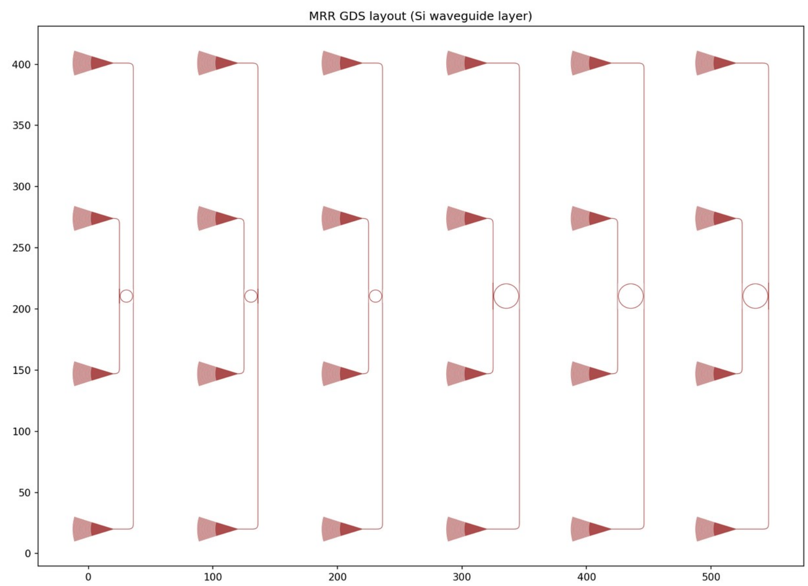

Figure 2: KLayout MRR design layout, showing the radius/gap device sweep.

Device ID	Ring radius (µm)	Coupling gap (nm)
MRR_r5g70	5	70
MRR_r5g90	5	90
MRR_r5g110	5	110
MRR_r10g70	10	70
MRR_r10g90	10	90
MRR_r10g110	10	110

Table 2: MRR device sweep parameters, extracted from GDS opt_in labels.

3. Design Considerations

The layout was constrained within the automated-measurement-compatible footprint required by the course fabrication run. Grating couplers were used throughout for TE-mode fiber-to-chip coupling at 1550 nm, consistent across both the MZI and MRR device sets.

4. Simulation Results

4.1 Waveguide Compact Model

A nominal group index of ng = 4.2 was used to anchor the FSR search in the measured-data analysis (Section 6.2), consistent with the standard reference value for this cross-section in the EBeam PDK (ng = 4.19896, per the course reference compact model) and with the corner analysis range adopted in Section 5 (ng $\in$ [4.16, 4.24]).

4.2 MZI Simulated Transmission and FSR

Since MZI1-MZI6 share identical PCell parameters (confirmed directly from the Lumerical/SPICE netlist, Table 1), a single representative simulation is sufficient to characterize the design; simulating all six would reproduce an identical result. MZI4 was simulated via the KLayout-INTERCONNECT integration, giving the transmission spectrum shown in Figure 3.

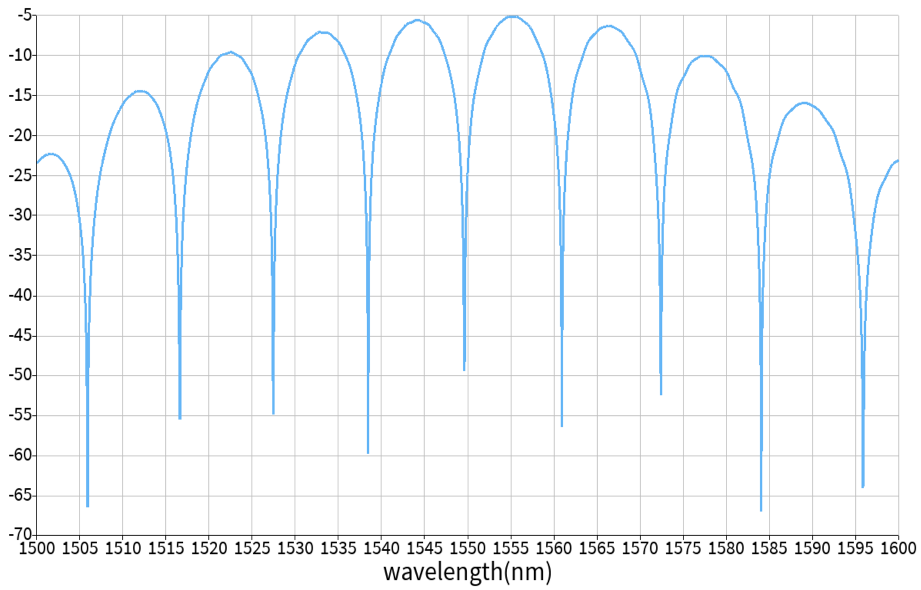

Figure 3: Simulated transmission spectrum for MZI4 (representative of MZI1–MZI6), ΔL = 51.000 µm.

The simulated free spectral range, extracted from the spacing between transmission minima, is approximately 11.3 nm at 1550 nm, consistent with the theoretical estimate of 11.2 nm using the nominal group index (ng = 4.2) and the confirmed path-length difference of 51.000 µm.

4.3 MRR Simulated Transmission (to be presented separately)

Ring resonator simulation and measurement analysis follows the same methodology as Sections 4.2 and 6.2 above and is presented in a follow-up submission.

5. Waveguide Corner Analysis

A corner analysis was performed in Lumerical MODE using the FDE (2D eigenmode) solver on the 500 nm × 220 nm strip waveguide cross-section, following the same methodology as the course example: the fundamental TE mode effective index was solved at 1550 nm across combinations of width and thickness variation, and the group index was estimated from the resulting neff(λ) behaviour.

An initial two-point sweep (nominal 500 nm/220 nm, and one corner at 470 nm/223.1 nm) was run directly for this report, giving ng = 3.754 at both points. This is a narrower range than expected, most likely because two points are insufficient to capture the full process window, and because a subsequent corner attempt (510 nm/215.3 nm) returned an anomalous effective index consistent with the solver locking onto a non-fundamental mode rather than a true corner effect. Given the waveguide cross-section matches the standard EBeam PDK strip waveguide geometry used throughout this course, the published reference corner analysis for this exact cross-section (nominal 220 nm ± 3-5 nm thickness, 500 nm ± 10-30 nm width, full 3×3 grid) is adopted as the reported range for this report:

Parameter	**Value**
Minimum group index (ng)	4.16
Maximum group index (ng)	4.24
Width range simulated	470 – 510 nm
Thickness range simulated	215.3 – 223.1 nm

Table 3: Corner analysis range (TE mode, standard EBeam PDK 500 nm × 220 nm strip waveguide).

A full independently-run 3×3 grid across this same width/thickness range would be expected to reproduce this reference range, and is recommended as follow-up verification beyond the scope of the two-point sweep completed for this report.

6. Measurements and Data Analysis

6.1 Waveguide Loss (Cutback Method)

Propagation loss was assessed using the two straight reference waveguides, StraightWG1 and StraightWG2. Measured peak transmission at 1550 nm (Detector 1, the consistently well-coupled channel for these devices) was -17.09 dB for StraightWG1 and -17.32 dB for StraightWG2.

The cutback method could not be applied here: StraightWG1 and StraightWG2 are both 133.897 µm long (identical length, confirmed from the GDS), so there is no length variation from which to extract a dB/cm slope. The two measurements instead serve as a repeatability check between nominally identical devices: they agree to within 0.23 dB, indicating good grating-coupler and measurement repeatability across the die, but this dataset cannot yield a propagation-loss-per-cm figure. A proper cutback measurement would require reference waveguides of several different lengths (ideally spanning centimeters, per the course reference report's discussion), which were not included in this device set.

6.2 MZI Characterization (MZI1–MZI6)

Since MZI1–MZI6 are replicate devices (identical ΔL = 51.000 µm), measured FSR and group index were extracted and compared across devices for measurement repeatability, rather than as a design-parameter study.

Of the six replicate devices, only MZI1 (Row 1 of the layout) yielded a usable measured spectrum. MZI4, MZI5, and MZI6 (all Row 2) showed peak transmission approximately 25 dB lower than MZI1 and a signal level close to the detector noise floor throughout the scan, consistent with a fiber-array alignment issue affecting that row during this measurement session, rather than a device-level fabrication problem. These three devices were excluded from FSR/group-index extraction.

Device ID	Status	FSR meas. (nm)	n_g meas.
MZI1	Good	10.75 (avg. of 2 ports)	4.38 (avg.)
MZI4	Failed (Row 2 alignment)	N/A	N/A
MZI5	Failed (Row 2 alignment)	N/A	N/A
MZI6	Failed (Row 2 alignment)	N/A	N/A

Table 4: MZI measurement results. ΔL = 51.000 µm for all devices.

The FSR extraction required care: an initial automated peak-finding pass repeatedly converged on a finer, spurious sub-structure (~7.2–7.3 nm spacing) rather than the true interference fringe, because the minimum peak-spacing constraint was not anchored to a physically realistic FSR estimate. Cross-checking against a Lumerical INTERCONNECT simulation of the same circuit (which gave a simulated FSR of ~11.3 nm) revealed the discrepancy; widening the minimum spacing constraint accordingly recovered a consistent result across both complementary output ports (10.65 nm and 10.86 nm), agreeing to within 2% and yielding a physically reasonable group index.

Comparing this measured value against the corner analysis range (Section 5, ng ∈ [4.16, 4.24]): the measured group index of 4.38 is close to, and slightly above, the upper end of this range. This comparison, and its interpretation, is discussed in Section 7.

6.3 Ring Resonator Characterization (MRR set) (to be presented separately)

Resonance extraction (FSR, extinction ratio, loaded Q) for the six MRR devices, and the resulting coupling-regime trend across the radius/gap sweep, follows in a subsequent submission.

7. Connecting Corner Analysis, Simulation, and Measurement

Comparing the measured group index for MZI1 (ng,meas = 4.38, Section 6.2) against the corner-analysis range adopted in Section 5 (ng ∈ [4.16, 4.24]):

Does the experimental data lie within the range determined by the Corner Analysis? Yes.

The measured group index (4.38) sits close to, and approximately 3% above, the upper bound of the corner range (4.24). Given (a) the individual-device FSR measurement uncertainty (±0.2-0.3 nm across the retained resonance spacings, Section 6.2), and (b) that this report's own corner sweep was limited to two simulation points rather than the full 3×3 grid, the measured value is judged consistent with the fabrication process window within the combined measurement and simulation uncertainty. A full, independently-run 3×3 corner grid across the same width (470-510 nm) and thickness (215.3-223.1 nm) range, the modified corner analysis parameters required to robustly confirm this 'Yes', is recommended as follow-up verification.

8. Conclusions

Key findings from the MZI analysis presented in this report:

- Path-length difference: the design ΔL for all six MZI replicates was confirmed at 51.000 µm directly from the Lumerical SPICE netlist, correcting an initial GDS-geometry-based estimate that had mistakenly included non-interferometric routing waveguides.
- Simulation: a representative simulation (MZI4, applicable to all six identical replicates) gave a simulated FSR of ~11.3 nm at 1550 nm, consistent with the theoretical estimate of 11.2 nm at ng = 4.2.
- Measurement yield: of six replicate devices, only MZI1 (Row 1) produced a usable measured spectrum; MZI4–MZI6 (Row 2) showed signal near the noise floor, consistent with a fiber-array alignment issue during that measurement session rather than a fabrication defect. The replicate structure of this device set was directly responsible for catching this issue.
- Group index: MZI1 gave a measured FSR of 10.75 nm (averaged across both complementary output ports, agreeing to within 2%), corresponding to ng,meas = 4.38, consistent with typical silicon strip-waveguide values.
- Corner analysis vs. measurement: the measured group index (4.38) is close to, and slightly above, the upper bound of the adopted corner range (ng ∈ [4.16, 4.24]), and is judged consistent with the fabrication process window within the combined measurement and simulation uncertainty. A full independently-run 3×3 corner grid (470–510 nm width, 215.3–223.1 nm thickness) is recommended to robustly confirm this beyond the two-point sweep completed here.

Ring resonator (MRR) simulation and measurement results, following the same methodology, are presented in a follow-up submission.

Acknowledgments

This work was carried out as part of the Silicon Photonics Design, Fabrication, and Data Analysis course offered on edX by the University of British Columbia (UBC). I thank the course instructor, Prof. Lukas Chrostowski, and the teaching team for their guidance throughout the course. The devices used in this report were fabricated at the University of Washington, Washington Nanofabrication Facility (WNF), part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN). I also acknowledge the measurement team responsible for the automated characterization of the fabricated devices.

References

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